



Internship Report

Deep Learning-Based Classification of Paper-Based Microscopic Images

1 Summary of Internship

This internship focused on applying a deep learning approach to classify paper-based microscopic images into cancerous and non-cancerous categories. The objective was to evaluate whether convolutional neural networks (CNNs) can detect subtle patterns in scanned or photographed paper-based tissue samples that may correlate with the presence of cancer.

The motivation behind this project was to explore the feasibility of using inexpensive, widely available imaging techniques (such as scanned printed histopathology slides) for preliminary medical screening. The scope included:

- Preprocessing of microscopic images.
- Designing and training a CNN model.
- Categorizing images into “NonCancer”, “Medium”, and “Cancer”.
- Analyzing model accuracy and prediction behavior.

2 Contribution

The core contribution involved building an end-to-end pipeline for binary classification using a CNN. The dataset consisted of labeled images, and each image was resized and normalized to standard dimensions for input into the neural network.

Image Preprocessing

Each image was loaded using OpenCV, resized to 128×128 pixels, and normalized by dividing pixel values by 255. This normalization allowed the network to handle RGB data on a consistent scale. The images retained their 3-channel (color) information, meaning the model learned not just from shape, but also from color distribution and intensity.

Model Architecture

The CNN consisted of:

- Two convolutional layers (with 32 and 64 filters) using ReLU activation.
- Two max-pooling layers to reduce spatial dimensions and retain key features.
- A flattening layer followed by a dense (fully connected) layer.
- A dropout layer with 50% rate to prevent overfitting.
- A final sigmoid layer that outputs probabilities for binary classification.

Training Strategy

The model was trained on the full dataset using 15 epochs and a validation split of 10%. Binary cross-entropy was used as the loss function, and accuracy was the primary evaluation metric. While the model trained well on the training data, validation accuracy remained low.

Prediction Mapping

Predicted probabilities were converted into three classes:

- **NonCancer** for scores below 0.5.
- **Cancer** for scores above 0.8.
- **Medium** for scores in between.

Accuracy and Pattern Recognition

The model achieved approximately 60–65% accuracy on the dataset. This accuracy reflects the model's ability to recognize color intensity and texture features present in the printed microscopic images. Since the input was color (RGB), the CNN likely picked up on variations in pink/purple hues that are characteristic of stained tissue slides used in pathology.

However, due to:

- Small dataset size,
- Potential label noise or ambiguity,
- Lack of augmentation or advanced preprocessing,
- Overfitting on training images,

the validation accuracy was significantly lower than the training accuracy — at times differing by 0.5 or more.

3 Results

The following image shows the training and validation accuracy curves over 15 epochs. As observed, training accuracy steadily improves, while validation accuracy plateaus and remains relatively lower, indicating overfitting:

The model achieved the following classification performance:

- **Overall Accuracy:** 60–65%.
- **NonCancer Accuracy:** 80-100%.
- **Cancer Accuracy:** 70-100
- **Medium Category:** Lower due to fewer samples and class imbalance. Introduced to capture uncertain predictions between 0.5 and 0.8.

The confusion in prediction categories arises mainly due to visual similarities in tissue patterns and low resolution of printed image inputs.

Image No	Predicted_Label	Actual_Label	Prediction_Score
image001.png	NonCancer	NonCancer	0.14
image002.png	NonCancer	NonCancer	0.10
image003.png	NonCancer	NonCancer	0.32
image004.png	NonCancer	NonCancer	0.15
image005.png	NonCancer	NonCancer	0.15
image006.png	NonCancer	NonCancer	0.38
image007.png	NonCancer	NonCancer	0.20
image008.png	NonCancer	Medium	0.24
image009.png	NonCancer	Medium	0.23
image010.png	Medium	Medium	0.72
image011.png	NonCancer	Medium	0.33
image012.png	Cancer	Medium	0.81
image013.png	NonCancer	Medium	0.33
image014.png	Cancer	Cancer	0.98
image015.png	Cancer	Cancer	0.96
image016.png	Cancer	Cancer	0.80
image017.png	Cancer	Cancer	0.86
image018.png	Medium	Cancer	0.71
image019.png	Cancer	Cancer	0.91
image020.png	NonCancer	Cancer	0.06

 Overall Prediction Accuracy: 65.00% (13/20)

 Per-Class Accuracy:

NonCancer : 100.00% (7/7)
Medium : 16.67% (1/6)
Cancer : 71.43% (5/7)

Figure 1: Output

4 Conclusions

The CNN model demonstrated moderate performance on classifying paper-based microscopic images, with training accuracy reaching around 60–65%. The model learned to recognize patterns such as texture, stain density, and color intensities, indicating that even low-cost image sources can provide learnable signals.

However, a noticeable gap between training and validation performance suggests the model overfit to the training set. Future work can improve this by:

- Increasing dataset size,
- Using data augmentation,
- Applying more advanced architectures or pre-trained models,
- Adding domain-specific features or stain normalization.

This work lays the groundwork for low-cost diagnostic support tools and highlights the potential of deep learning in resource-constrained environments.

References

@article url = <https://www.sciencedirect.com/science/article/pii/S2772906024005430>